

Task b:

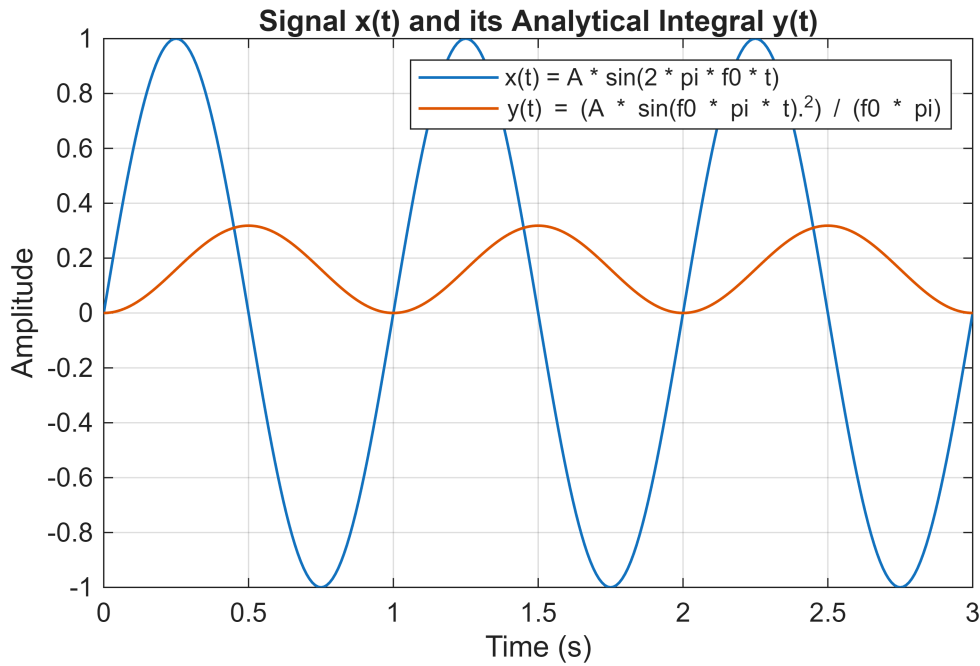
```
syms A f0 t pi
xt = A*sin(2*pi*f0*t);
yt = int(xt,'t',[0,t])
```

yt =

$$\frac{A \sin(f_0 \pi t)^2}{f_0 \pi}$$

Task C:

```
clear;
close all;
clc;
t = 0:0.01:3;
A = 1;
f0 = 1;
xt = A * sin(2 * pi * f0 * t);
yt = (A * sin(f0 * pi * t).^2) / (f0 * pi);
figure('Name', 'Task (c): Analytical Plot');
plot(t, xt, 'linewidth', 1);
hold on;
plot(t, yt, 'linewidth', 1);
grid on;
title('Signal x(t) and its Analytical Integral y(t)');
xlabel('Time (s)');
ylabel('Amplitude');
legend('x(t) = A * sin(2 * pi * f0 * t)', 'y(t) = (A * sin(f0 * pi * t).^2) / (f0 * pi)');
```



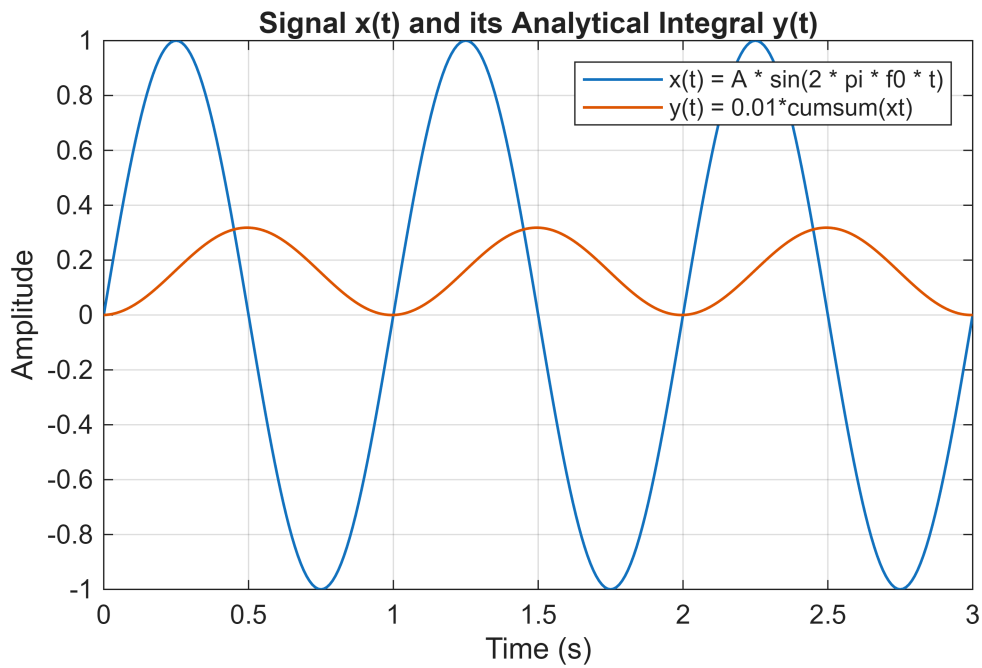
Task D:

```
a=[1,2,3,4,5];
b= cumsum(a)
```

```
b = 1x5
     1     3     6    10    15
```

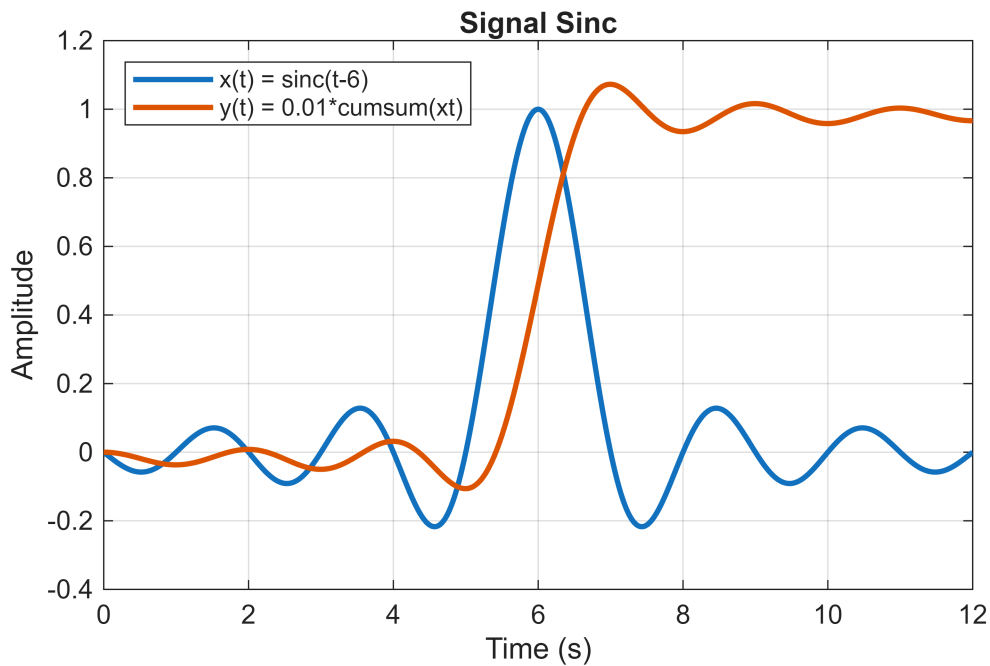
Task E:

```
clear;
close all;
clc;
t = 0:0.01:3;
A = 1;
f0 = 1;
xt = A * sin(2 * pi * f0 * t);
yt = 0.01*cumsum(xt);
figure('Name', 'Task (c): Analytical Plot');
plot(t, xt, 'linewidth', 1);
hold on;
plot(t, yt, 'linewidth', 1);
grid on;
title('Signal x(t) and its Analytical Integral y(t)');
xlabel('Time (s)');
ylabel('Amplitude');
legend('x(t) = A * sin(2 * pi * f0 * t)', 'y(t) = 0.01*cumsum(xt)');
```



Task H:

```
clear; close all; clc;
t = 0:0.01:12;
xt = sinc(t-6);
yt = 0.01*cumsum(xt);
figure('Name', 'Task (H): Sinc');
plot(t, xt, 'linewidth', 2);
hold on;
plot(t, yt, 'linewidth', 2);
grid on;
xlim([0 12]);
title('Signal Sinc');
xlabel('Time (s)');
ylabel('Amplitude');
legend('x(t) = sinc(t-6)', 'y(t) = 0.01*cumsum(xt)', 'Location', 'northwest');
```



Task I:

```
clear; close all; clc;
a=[1,2,3,4,5]
```

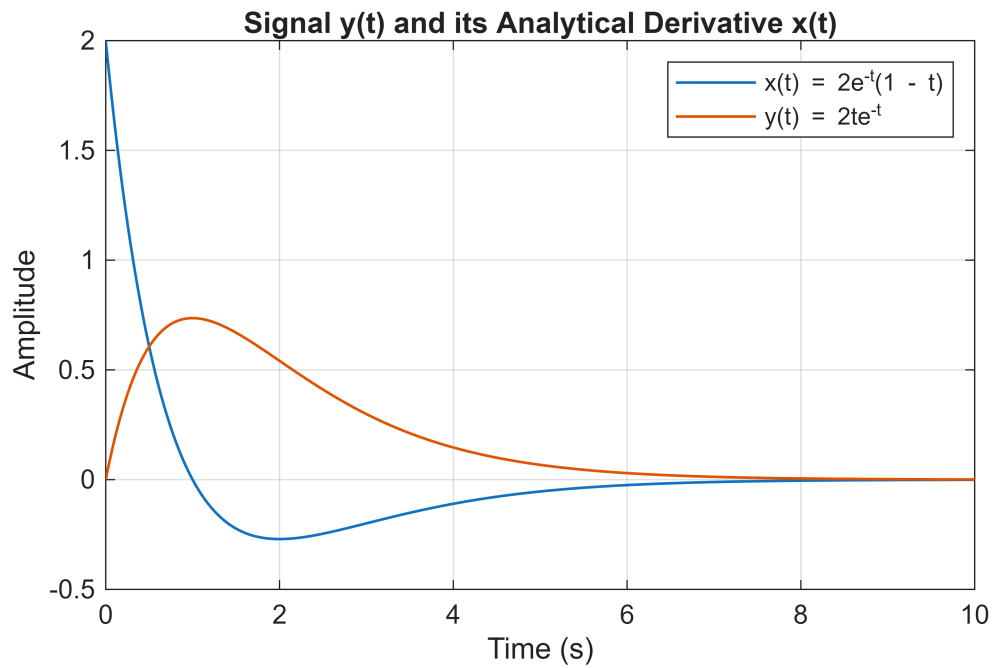
```
a = 1x5
     1     2     3     4     5
```

```
b=diff(a)
```

```
b = 1x4
     1     1     1     1
```

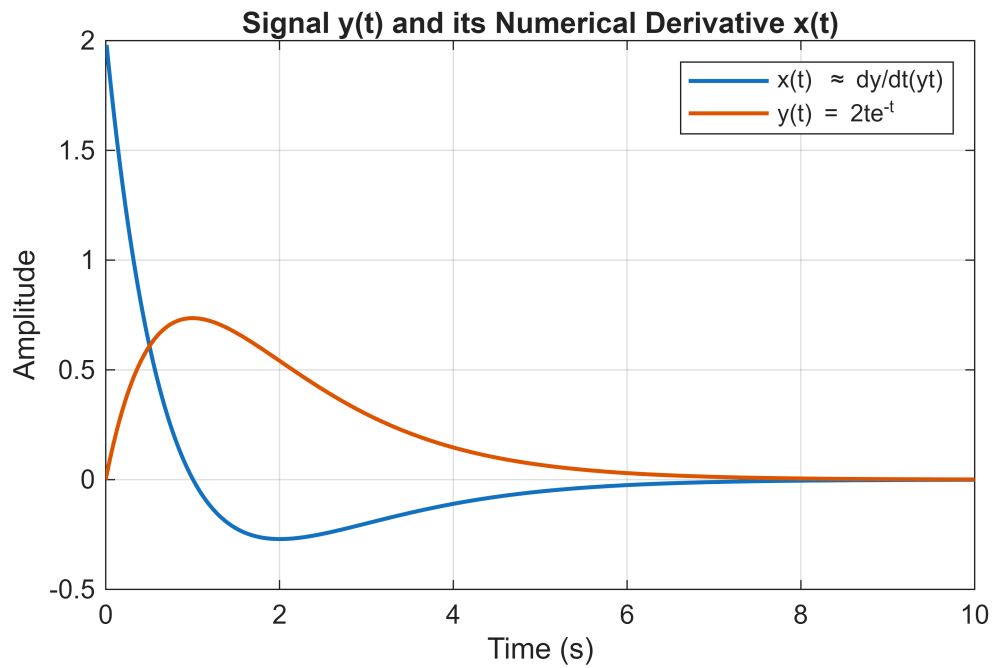
Task K:

```
clear;
close all;
clc;
t = 0:0.01:10;
xt = 2 * exp(-t) .* (1 - t);
yt = 2 * t .* exp(-t);
figure('Name', 'Task (K): Numerical Differentiation of a Signal');
plot(t, xt, 'linewidth', 1);
hold on;
plot(t, yt, 'linewidth', 1);
grid on;
title('Signal y(t) and its Analytical Derivative x(t)');
xlabel('Time (s)');
ylabel('Amplitude');
legend('x(t) = 2e^{-t}(1 - t)', 'y(t) = 2te^{-t}');
```



Task L:

```
clear;close all;clc;
dt = 0.01;
t = 0:dt:10;
yt = 2 * t .* exp(-t);
xt = diff(yt) / dt;
xt = [NaN, xt];
figure('Name', 'Task (L): Numerical Differentiation');
plot(t, xt, 'linewidth', 1.5);
hold on;
plot(t, yt, 'linewidth', 1.5);
grid on;
title('Signal  $y(t)$  and its Numerical Derivative  $x(t)$ ');
xlabel('Time (s)');
ylabel('Amplitude');
legend('x(t) \approx dy/dt(yt)', 'y(t) = 2te^{-t}');
```



Task M:

```
clear;close all;clc;
dt = 0.01;
t = 0:dt:10;
yt = sinc(t-6);
xt = diff(yt) / dt;
xt = [NaN, xt];
figure('Name', 'Task (L): Numerical Differentiation');
plot(t, xt, 'linewidth', 1.5);
hold on;
plot(t, yt, 'linewidth', 1.5);
grid on;
title('Signal y(t) and its Numerical Derivative x(t)');
xlabel('Time (s)');
ylabel('Amplitude');
legend('x(t) \approx dy/dt(yt)', 'y(t) = sinc(t-6)');
```

